

AMMAR AHMED

Mechanical Design | Mechatronics | Aerospace Systems

Portfolio focused on CAD-to-hardware development, UAV structures, rocket and launch vehicle concepts, mechatronic integration, FEA/CFD validation, thermal hardware, and rapid prototyping.

01
Concept and
requirements

02
CAD and simulation

03
Prototype build

04
Test and iterate

CONTACT

Erlangen, Germany | +49 1556 0360166

ammarahmed.00748@gmail.com

linkedin.com/in/ammar-ahmed11/ | github.com/EnggAmmar

enggammar.github.io/Portfolio_ammар/



Mechanical, Mechatronics and Aerospace Capability Map

A portfolio map for roles where mechanical design decisions must survive manufacturing, integration, testing, and simulation review.

Mechanical Design

CAD, assemblies, interfaces, packaging, mechanisms, design for manufacturing.

Aerospace Systems

UAV structures, SLV concepts, rocket engine layout, aerodynamics.

Mechatronics

Sensors, actuators, motors, CAN bench validation, robot mechanisms.

Simulation

FEA, CFD, thermal studies, parametric analysis, engineering dashboards.

TOOLS AND METHODS

SolidWorks

Siemens NX

CATIA v6

ANSYS Mechanical

ANSYS Fluent

MATLAB

Python

FDM 3D Printing

CAN Testbench

Power BI

FastAPI

React

EXPERIENCE EVIDENCE

WS Audiology | Device Commissioning Engineer

Subsystem and system-level validation, test documentation, data dashboards, component records.

NASTP | Research Officer

UAV prototype manufacturing and assembly, demonstration drones, electronics and landing gear integration, composite direction support.

Aero-Vision Technologies | Mechanical Design Engineer

Foldable fixed-wing UAV structures, FDM prototypes, CFD support, mechanical-electronic integration.

FAU MSc Electromobility

Mechatronics components and systems, robot mechanisms, electric drives, signal processing and ML for engineers.

UAV Airframe Design, Fabrication and Integration



Assembled fixed-wing foldable UAV prototypes prepared for review.



Meshed airframe geometry for CFD preparation.



FDM UAV structure before system assembly.



UAV wing and structural fabrication reference.

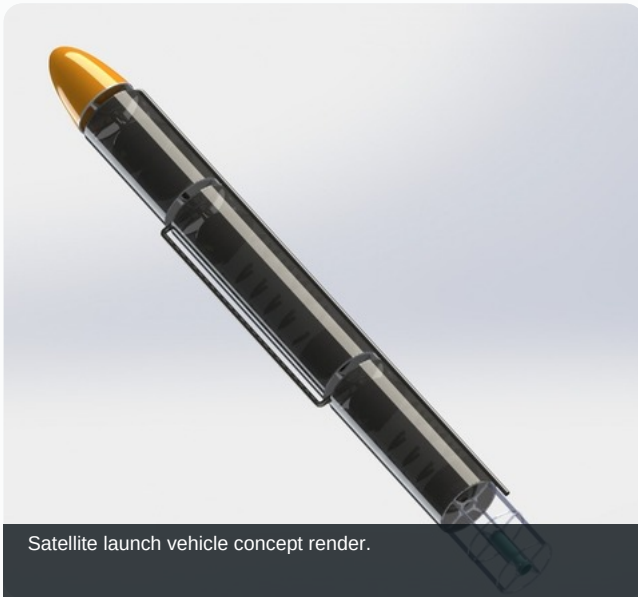
MECHANICAL AND AEROSPACE CONTRIBUTION

- Designed and supported foldable fixed-wing UAV structures using CAD-driven mechanical development.
- Converted CAD geometry into FDM printed prototype parts for fit checks and fast design iteration.
- Integrated structures with electronics, propellers, actuators, landing gear, and sensor-related hardware.
- Supported aerodynamic validation through CFD/mesh preparation and prototype refinement.
- Prepared demonstration-ready UAV models for stakeholder review and engineering decision-making.

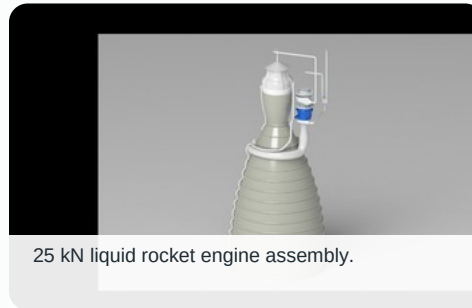
BUILD LOOP



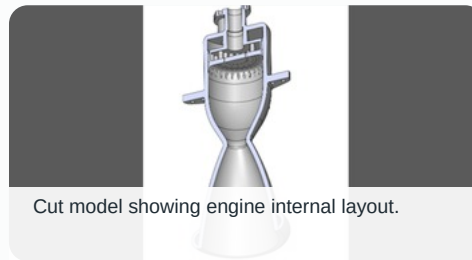
Satellite Launch Vehicle and 25 kN Rocket Engine Concept



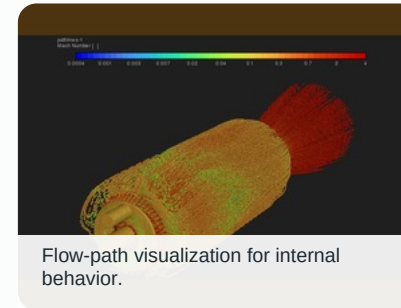
Satellite launch vehicle concept render.



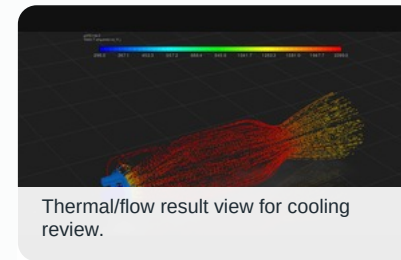
25 kN liquid rocket engine assembly.



Cut model showing engine internal layout.



Flow-path visualization for internal behavior.



Thermal/flow result view for cooling review.

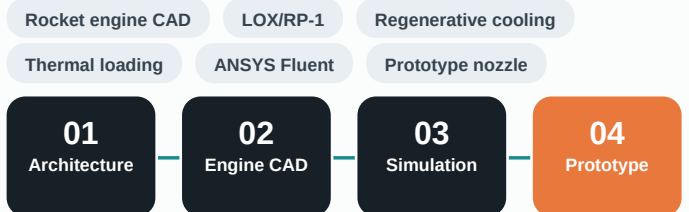


3D printed nozzle prototype for physical review.

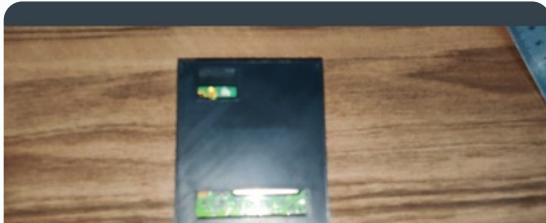
SYSTEM DESIGN SCOPE

- Conceptual launch vehicle and LOX/RP-1 engine architecture with propellant tanks, piping interfaces, and assembly layout.
- Engine CAD including injector, chamber, nozzle, internal routing, and communication-friendly cut views.
- ANSYS-supported stress and thermal review for tanks, nozzle regions, and cooling-related design decisions.
- Prototype nozzle printed for physical inspection, geometry communication, and manufacturability review.

ENGINEERING EMPHASIS



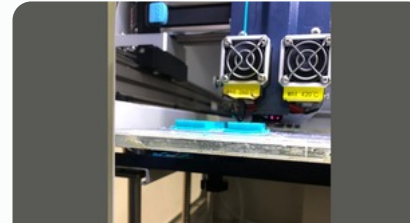
Mechanical Prototyping and Thermal Hardware



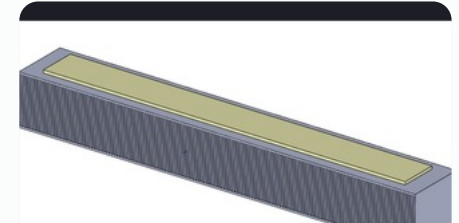
FDM printer setup for prototype fabrication.



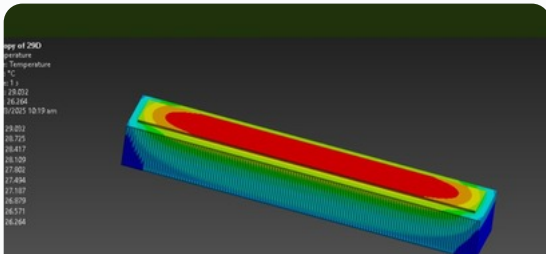
Printer operation and build setup.



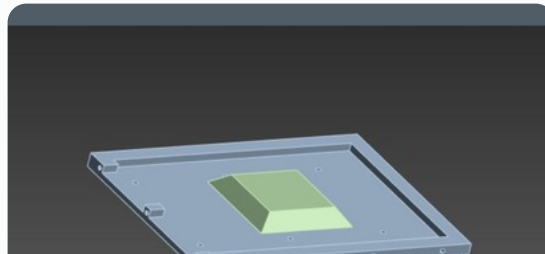
Printed model for fit validation.



Multi-PCB heat-sink concept.



Thermal hardware packaging view.



Thermal conduction and geometry study.

MECHANICAL INTERFACE AND ADDITIVE MANUFACTURING

- Prepared CAD models for FDM manufacturing and rapid design validation.
- Operated 3D printing workflows for PLA/ABS engineering prototype parts.
- Performed physical fit checks, alignment review, and assembly refinement.
- Used prototype feedback to improve mechanical interfaces and part usability.

THERMAL DESIGN EVIDENCE

- Developed compact electronics cooling concepts for multi-PCB layouts.
- Connected mechanical packaging constraints with simulation-informed geometry choices.
- Used thermal visualization and conduction studies to communicate design behavior.

CAD-to-part

FDM

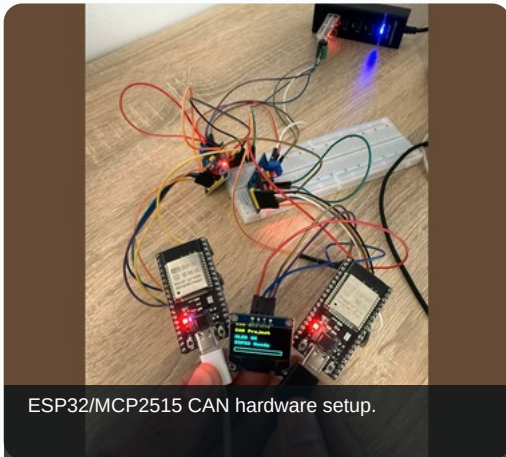
Fit checks

Heat sinks

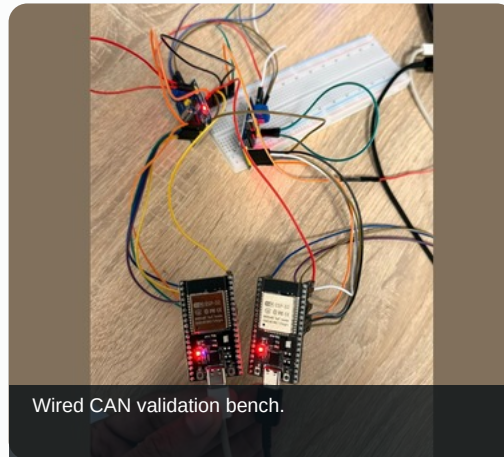
Electronics cooling

Thermal simulation

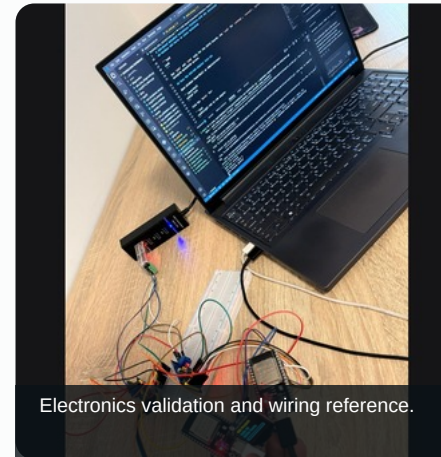
Mechatronic Systems, Controls and Validation



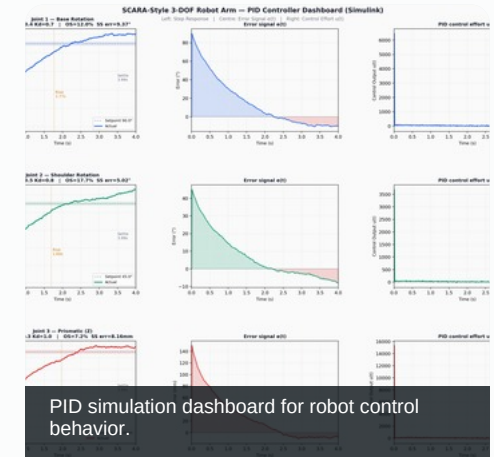
ESP32/MCP2515 CAN hardware setup.



Wired CAN validation bench.



Electronics validation and wiring reference.



PID simulation dashboard for robot control behavior.

HARDWARE INTEGRATION AND VALIDATION

- Built a repeatable electronics testbench using ESP32 and MCP2515 CAN hardware.
- Structured validation with wiring checks, message flow, signal behavior, and PASS/FAIL style outputs.
- Connected mechanical prototype thinking with embedded hardware setup and troubleshooting.
- Applied test documentation practices from device commissioning to engineering validation workflows.

ROBOT MECHANISM AND CONTROL DIRECTION

- SCARA-style RRP robot arm concept linking mechanism layout with control response behavior.
- Mechatronics coursework foundation in robot mechanisms, multibody electric drives, and signal processing.
- Control-loop visualization used to communicate system response and tuning direction.

CAN

ESP32

MCP2515

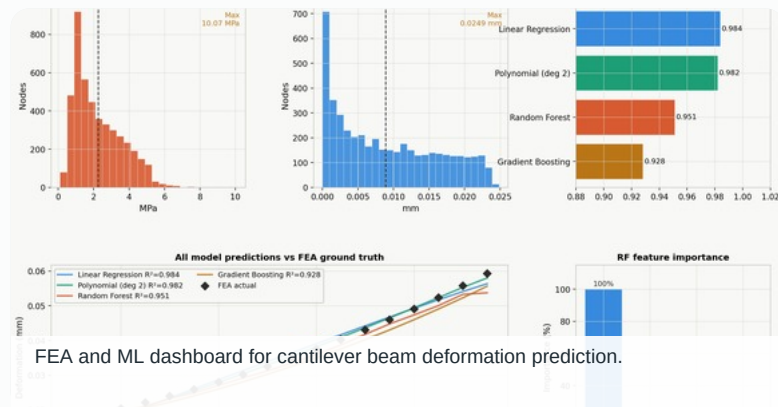
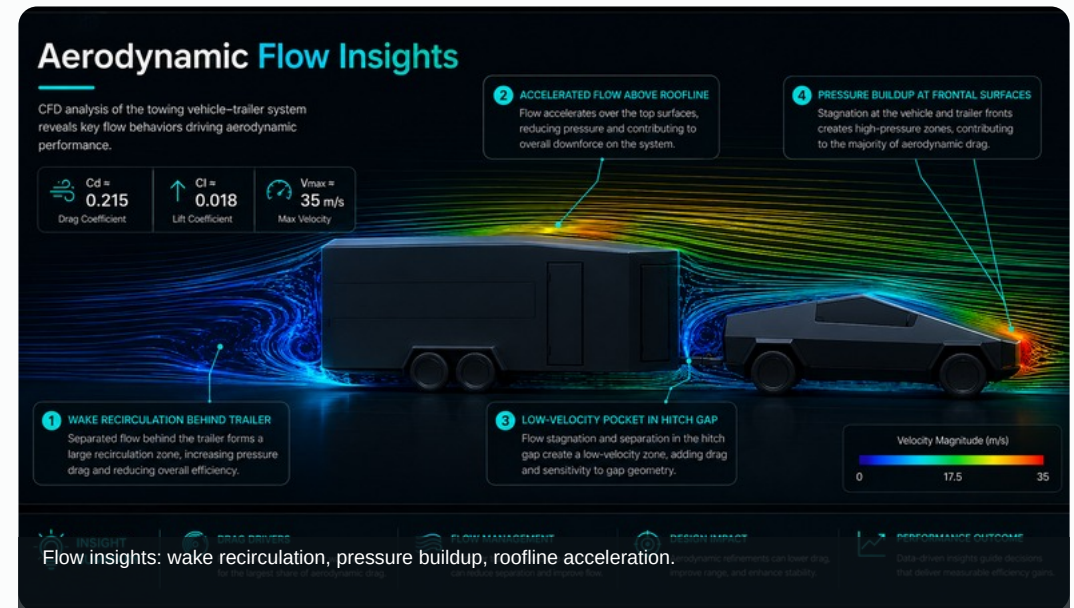
Sensors

Actuators

PID

Robot mechanisms

Simulation-Driven Design and Engineering Analytics



CFD AND FEA WORKFLOW

- Used ANSYS Fluent workflows for geometry preparation, meshing, solver setup, convergence review, and post-processing.
- Evaluated pressure distribution, velocity fields, wake behavior, and force coefficient trends for aerodynamic decisions.
- Performed parametric FEA and machine-learning analysis for deformation prediction and design insight extraction.
- Built engineering dashboards that turn simulation outputs into interpretable design evidence.

ANSYS Fluent

CFD

FEA

Parametric studies

Engineering analytics

Python

scikit-learn

READY FOR HARDWARE-CENTRIC ENGINEERING ROLES

Best fit: mechanical design, mechatronic systems, aerospace prototyping, UAV hardware, test and validation, rapid manufacturing, and simulation-supported product development.

CORE VALUE

- Turns CAD concepts into manufacturable prototype hardware.
- Connects mechanical assemblies with electronics, sensors, actuators, and test workflows.
- Uses FEA/CFD and dashboards to support design decisions with evidence.
- Works across aerospace, mechanical, thermal, and mechatronic system boundaries.

Ammar Ahmed

Erlangen, Germany

+49 1556 0360166

ammarahmed.00748@gmail.com

[linkedin.com/in/ammahmed11/](https://www.linkedin.com/in/ammahmed11/)

github.com/EnggAmmar